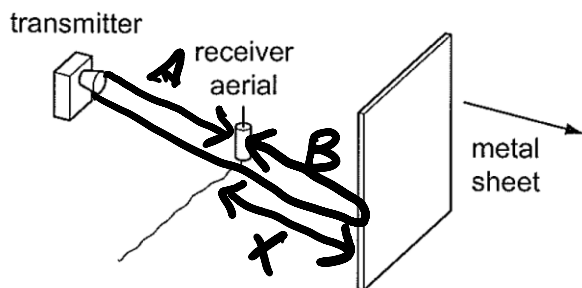


unit area) is 12 times of its original value.

18

B

Method 1:



At the receiver, the incoming wave (path A) meet the reflected wave (path B). The resultant displacement is the vector sum of individual displacement of the two waves.

When path difference $|B - A| = 2x = (n + \frac{1}{2})\lambda$, the waves arrive in phase, leading to constructive interference, so detected voltage is maximum. This happens at distance moved = 30 mm, from graph provided.

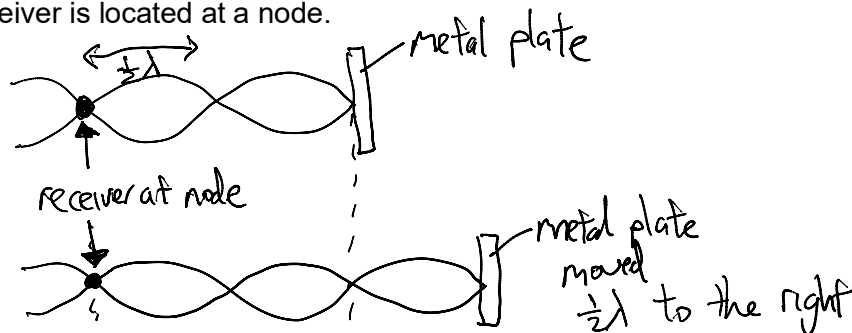
When metal sheet has moved another 60 mm (at distance moved = 90 mm), the voltage detected is maximum again, so it is another constructive interference, with path $|B - A| = 2(x + 60) = 2x + 120 = (n + 1 + \frac{1}{2})\lambda$.

Hence, $\lambda = 120$ mm

Method 2:

The incoming wave from transmitter will undergo phase change of π rad upon reflection at the metal plate. The incoming and reflected waves will form a stationary wave, with a node at the metal plate.

When distance moved = 0, a minimum voltage was detected, showing that the receiver is located at a node.



	When metal plate moved by $\frac{1}{2}\lambda$ to the right, the stationary receiver will be at a node again, detecting a minimum voltage. This happens when distance moved = 60 mm. Hence, $\frac{1}{2} \lambda = 60 \text{ mm}$, so $\lambda = 120 \text{ mm}$
19	D